

Programming Techniques – Using C Language

Objective: The purpose of this lab is to become familiar with Function.

Functions

1. Type and execute the given program and get an idea about the operation of a function.

```
#include <stdio.h>
int cube(int c); /* This is a function prototype */
int main(void){
    int x = 10, y = 7;
    int v;
    printf("X = %d\n",x);
    v = cube(x);    /* Call the cube function using x as argument */
    printf("Now X = %d\n\n",v);
    printf("Y = %d\n",y);
    v = cube(y);    /* Call the cube function using y as argument */
    printf("Now Y = %d\n\n",v);
    return 0;
}

int cube(int c){
    return (c*c*c);
}
```

- a. Modify above program to read values to variables x and y from the keyboard.
2. Type and execute the following program. This program is for get an idea about void functions

```
#include <stdio.h>
void square(int num);

int main() {
    square(5);
    return 0;
}

void square(int num) {
    // Function to calculate the square of a number.
    int result = num * num;
    printf("%d\n", result);
}
```

3. Write a program in C to check if a given number is even or odd using the function.
4. Write a C program to find the largest number from the given three numbers. Write a C function that takes three integers as arguments and returns the value of the largest one.
5. Write a C program to calculate the factorial value using separate function call factorial (int n) to calculate factorial and return the factorial value of given integer n.
6. Write a C Program to enter any character from user and if that character is capital letter then display it with small letter on screen and if enter character is small letter then display with Capital letter.
 - a. Read the character inside the main method.
 - b. Define a function to check and convert the case of the given character. This function should return the converted character.
7. Write a C program to input 10 integers to an array and calculate the summation of the array elements using given function call **sum (int arr[], int n)** to calculate the sum.
 - a. Define a function which returns the summation of the ten integers (**int sum(int arr[], int n)**) which has two arguments for passing the array of integers and the size of the array.
8. Write a C program to input the first name, middle name and last name and then output the fullname. Write a void function call **Fullname** which have three arguments (**fname, mname, lname**) create the full name and display the full name.
9. Write a C program to reverse the given string.
 - a. Define a separate void function to handle the reversing process.